

Measuring Human and System Signals in Voice-Centric Customer Service Agents

Executive summary

The best-supported architecture for a production voice customer-service agent is **not** “send the WAV to one emotion model and trust the answer.” It is a **multilayer measurement system** that preserves synchronized views of the interaction: raw or policy-approved audio, timestamped transcript, deterministic conversational timing, low-level acoustic/prosodic measurements, learned audio representations, agent/system telemetry, and business outcomes. Recent customer-service research supports combining linguistic and acoustic information, while systematic reviews show that individual acoustic correlates are often nonspecific and context-dependent. ¹

Modern native-audio models can consume speech directly and react to intonation, pace, interruptions, and semantic context. OpenAI, Google, and Microsoft now expose low-latency voice-agent architectures that reduce or eliminate the traditional STT→LLM→TTS cascade. Google explicitly documents native-audio processing, interruptions, and configurable affective dialogue; Microsoft Voice Live combines recognition, generative AI, and synthesis behind one interface; OpenAI's realtime voice models similarly support direct, streaming voice interaction. ² **However, native audio reasoning should complement rather than replace an independent analytics/telemetry plane.** End-to-end model judgments are harder to calibrate, version, audit, compare longitudinally, and reproduce than explicit measurements.

The scientific literature supports a narrower interpretation of “emotion detection” than product marketing often implies. A 2025 systematic review covering 38 articles and 39 studies found anger commonly associated with higher F0, greater intensity and faster speech; stress with higher F0/intensity and shorter duration; but it found no consistent acoustic pattern for fear or anxiety. A systematic review of perceived vocal trustworthiness likewise found that multivariate, contextual analysis is more informative than isolated cues such as pitch, whose findings are often inconclusive. ³ Therefore, production systems should represent affect as **probabilistic, contextual signals**, preferably continuous valence/arousal plus task-specific states such as “frustration evidence,” rather than claiming to know a customer's true emotion.

The industry has converged on a pragmatic set of metrics. Amazon Connect Contact Lens exposes speaker-turn sentiment, talk/non-talk time, loudness, response-related metrics and PII handling; Google Customer Experience Insights analyzes sentiment, interruptions, silence, entities and intents; Genesys exposes sentiment, topics, transcript analytics and agent-empathy analysis. These observable interaction metrics are much better established operationally than deep psychological inference. ⁴

My recommended production strategy is:

Decision	Recommendation
Primary unit of analysis	Use turns plus rolling time windows , not entire-call labels. Maintain 20–50 ms acoustic frames internally, aggregate acoustic measures over roughly 1–5 s, interpret semantic/affective state at turn level, and compute trajectories over 30–120 s and whole calls.
Essential signals	Transcript + timestamps/confidence; VAD; turn boundaries; gaps, overlap and interruption; silence; response latency; system/tool/TTS latency; intent/topic; repair/repetition; outcome and explicit customer-feedback signals.
Recommended acoustic signals	Speaker-normalized F0, F0 range, intensity, speech/articulation rate, pause statistics and a compact eGeMAPS-style feature set. GeMAPS/eGeMAPS were designed as standardized, interpretable affective-voice feature sets. ⁵
Recommended learned signals	Emotion-specific/self-supervised embeddings, calibrated valence/arousal or negative-affect probability, and multimodal fusion with text. emotion2vec is an example of a representation model that improved emotion-related tasks across ten languages, although deployment still requires domain validation. ⁶
Avoid as primary truth	Acoustic-only trust, compliance, deception, personality, anxiety, diagnosis, or employee emotional-state scoring. Evidence is too context-dependent for such interpretations. ⁷
Real-time architecture	Keep the conversational serving path latency-critical; run lightweight VAD/timing/prosody continuously and semantic analysis incrementally. Push expensive embedding, full eGeMAPS, diarization refinement, quality scoring and longitudinal analysis to an asynchronous lane. Spoken-dialogue research specifically identifies timing and runtime ASR latency as central bottlenecks. ⁸
Governance	Treat emotion estimates as measurements with uncertainty, not facts. Validate separately by locale, channel and customer population; retain calibration and model versions; implement abstention; minimize storage of raw audio and speaker embeddings. NIST AI RMF emphasizes governance, measurement and management across the lifecycle, while ISO/IEC 42001 establishes an AI-management-system framework. ⁹
European deployments	Carefully distinguish customer analytics from employee analytics. The EU AI Act defines emotion-recognition systems and prohibits AI emotion inference in workplace and educational settings except specified medical/safety uses; it also imposes transparency duties for permitted emotion-recognition deployments. This is especially consequential if the system scores human agents , not just customers. ¹⁰

The most important design principle is consequently:

Instrument what happened first; infer why it happened second.

Response delay, interruption, silence, escalation, repetition, failed tools and resolution are observable. “Frustrated,” “trusting,” “engaged,” or “likely to comply” are latent interpretations and should require progressively stronger evidence and validation.

Evidence base and what the science and industry support

The research landscape from approximately 2016–2026 shows a shift from handcrafted acoustic features toward self-supervised representations and native-audio foundation models, but it has **not** removed the fundamental problems of label ambiguity, acted-versus-naturalistic domain shift, channel effects, demographic/cultural variation and contextual dependence. A particularly relevant 2026 customer-care study combines linguistic and acoustic information at each turn and explicitly notes that scarcity of realistic datasets can degrade industrial performance. A 2025 study focusing on customer-service calls similarly highlights performance gaps between acted and naturalistic emotion data. ¹¹

Selected scientific literature

Period / work	Contribution	Production implication
Eyben et al., GeMAPS, 2016	Introduced a deliberately compact standardized acoustic parameter set for voice research and affective computing. ¹²	Prefer standardized feature definitions to arbitrary hundreds/thousands of acoustic variables when interpretability matters.
Kamiloğlu et al., systematic review, 2020	Reviewed vocal production of positive emotions; acoustic patterns vary among positive-emotion families rather than reducing cleanly to “positive voice.” ¹³	Do not treat pitch or loudness as universal valence detectors.
Parra-Gallego et al., 2022	Studied speech emotion classification and customer-satisfaction evaluation in realistic acoustic environments. ¹⁴	Customer-service evaluation should use real acoustic conditions and direct satisfaction outcomes.
Liesenfeld & Dingemanse, timing-bottleneck analysis, 2023	Argues that ASR evaluation for conversational systems must include timing, diarization, overlap and coverage in addition to WER; runtime latency obstructs fluid interaction. ¹⁵	Treat timing quality as a first-class model metric, not just transport telemetry.
Niebuhr & colleagues, codec study, 2023	Found codec-specific distortions across all tested prosodic parameters, with mean automatic measurements deviating by as much as roughly 20% in their experimental conditions. ¹⁶	Never compare raw prosody across PSTN/WebRTC/codecs without channel controls and normalization.

Period / work	Contribution	Production implication
Ma et al., emotion2vec, 2024	Self-supervised emotion representation with utterance- and frame-level objectives; reported improvements across ten languages and several emotion-related tasks. ⁶	Learned embeddings are useful secondary features, especially after domain fine-tuning, but remain opaque compared with interpretable prosody.
Speech-demographic bias study, 2024	Found statistically significant demographic differences in acoustic features, labels, and model performance in speech-based behavioral/mental-health prediction. ¹⁷	Audit customer-service models for subgroup and locale performance; normalization does not eliminate all bias.
Schewski et al., systematic review, 2025	Across 39 studies, prosodic features were the most studied; anger showed relatively recurrent F0/intensity/rate relationships while fear/anxiety did not have consistent acoustic signatures. ¹⁸	Acoustic affect should be probabilistic and multimodal; don't map one feature to one state.
Maltezou- Papastyliaou et al., systematic review, 2025	Acoustic properties influence perceived trustworthiness, but isolated pitch findings are inconclusive; multivariate/contextual analysis performs conceptually better. ¹⁹	Measure customer trust directly when it matters; do not equate low/high pitch with trust.
Castillo-López et al., turn-taking survey, 2025	Reports substantial remaining challenges in spoken-agent turn-taking and a lack of established benchmarks; reviews linguistic, timing and speech cues. ²⁰	End-of-turn detection, barge-in and gap timing deserve independent evaluation.
Customer-service naturalism work, 2025	Examines acted versus naturalistic speech-emotion performance gaps in customer-service calls. ²¹	Avoid reporting IEMOCAP/acted-corpus scores as expected contact-center performance.
Gangopadhyay & Mehrabani, 2026	Turn-level negative-emotion detection on real enterprise customer-support data using both linguistic and acoustic features, designed for realtime operation. ²²	A practical production pattern is turn-level multimodal detection feeding a lightweight real-time policy.

Several conclusions recur across this literature. First, **prosody is useful but nonspecific**: elevated F0 and intensity can accompany anger, stress, cognitive load, emphasis, excitement or deliberate speaking style. ²³ Second, **channel and speaker variation are large enough to demand normalization**; speech compression alone can distort acoustic measurements materially. ¹⁶ Third, **subjective states do not have perfectly objective ground truth**, and speech classifiers can be overconfident, making calibration and abstention important for deployment. ²⁴

What contact-center vendors actually measure

Commercial contact-center platforms are useful evidence of what has proven operationally measurable at scale, although vendor claims about predictive business value should not be confused with independent scientific validation.

Platform	Documented analytics	What this says about industry practice
Amazon Connect Contact Lens	Speaker-turn sentiment; positive/neutral/negative aggregation; talk and non-talk time; silence; loudness; response-related metrics; issue/category analysis; sensitive-data handling. ²⁵	Strong emphasis on transcript semantics plus observable interaction timing and audio statistics.
Google Customer Experience Insights / Quality AI	Sentiment, interruptions, silence/hold, entities, intents, highlights and quality analysis; Google's quality guidance recommends sufficiently large conversation samples and supports PII-redacted transcripts. ²⁶	QA is increasingly whole-conversation and multimodal, but driven by observable criteria rather than unconstrained psychological diagnosis.
Genesys Speech and Text Analytics	Sentiment, topics, categories, transcription and empathy-related analytics. Genesys describes empathy analysis using voice transcripts. ²⁷	"Empathy" products may be primarily linguistic rather than physiological/acoustic measurements; inspect implementation before interpreting the label.
NICE	Commercial emotion/sentiment tooling describes tone, pace, pitch and word choice among analyzed signals. ²⁸	Demonstrates demand for multimodal analytics, but marketing statements should not establish scientific validity.
Microsoft customer-service analytics	Real-time customer sentiment and historical voice analytics, with explicit reminders that deployers remain responsible for applicable monitoring, recording and storage laws. ²⁹	Sentiment is operationally useful, but governance and deployment context remain the customer's responsibility.

ISO 18295-1 establishes a service-requirements framework for customer contact centers and explicitly encompasses performance metrics where appropriate, while ISO/IEC 42001 and ISO/IEC 23894 provide broader AI management and risk-management structures. These standards support a **KPI + governance + continuous-monitoring** approach rather than reliance on a single "AI quality score." ³⁰

Measurement feature catalog and psychological mappings

For the tables below, **measurement reliability** means how reproducibly the signal itself can be extracted—not whether it uniquely reveals a psychological state.

High means predominantly objective given adequate audio and instrumentation. **Moderate** means robust enough for production after channel/speaker/domain normalization. **Low-moderate** means materially affected by signal quality, model domain, demographic/language variation, subjective labels or all four. This rating is my synthesis of the literature and tooling, rather than a standardized published scale. The recommended feature definitions follow established speech-processing capabilities in GeMAPS/eGeMAPS, openSMILE, Praat and librosa. ³¹

Feature catalog

Dimension / feature	Definition and extraction	Typical realtime window / aggregation	Reliability	Main confounds	Example tools
ASR words + timestamps	Recognized lexical sequence with word/segment start/end times and confidence where available.	Incremental partials; finalize per turn; conversation-level corrected transcript offline.	High-moderate	Accent, language, crosstalk, noise, jargon, names, code-switching.	Whisper; Azure Speech; NVIDIA Riva; commercial cloud STT. Whisper was trained as a robust multilingual general-purpose system; Azure exposes realtime and batch modes; Riva provides streaming ASR. ³²
Lexical sentiment / valence	Probability or continuous score for positive/neutral/negative semantic content.	Per utterance/turn; contextual smoothing over ~2–6 turns; beginning/end/whole-call aggregates.	Moderate	Sarcasm, negation, ASR errors, domain vocabulary, quoted speech, cultural style.	Domain classifier, LLM, AWS Contact Lens, Google CX Insights, Genesys. ³³
Intent / topic / dialogue act	Task intent and conversational function such as request, confirmation, rejection, clarification or acknowledgment.	Per turn with conversation context.	High-moderate after domain training	Novel intents, multi-intent turns, ASR errors.	LLM/NLU classifiers; Google CX Insights; contact-center NLP. ³⁴

Dimension / feature	Definition and extraction	Typical realtime window / aggregation	Reliability	Main confounds	Example tools
Entities / PII	Names, account identifiers, addresses, payment/health information and other domain entities.	Token/turn; redact as close to ingestion as technically feasible.	Moderate-high	ASR misspellings, context ambiguity, novel entity formats.	Cloud DLP/ contact-center redaction, NER, LLM plus deterministic rules. AWS and Google document sensitive-data/ PII workflows. 35
Displaying Entities for verification during the interaction should resolve this					
Repair / repetition language	"Sorry?", corrections, restarts, repeated questions, reformulation and explicit misunderstanding.	Per turn; rolling count over 30–120 s and entire interaction.	High-moderate	Natural emphasis/ repetition; ASR duplicated segments.	Transcript rules + dialogue-act classifier/LLM.
Hedges / uncertainty language	"Maybe," "I think," "not sure," modal uncertainty and tentative wording.	Per turn; contextual trajectory.	Moderate	Politeness conventions, language/ culture, script requirements.	Text classifier/ LLM; combine with acoustic evidence rather than interpreting alone. Vocal confidence cues themselves are context/ language dependent. 36
F0 level	Fundamental frequency of voiced speech; usually median rather than mean.	20–50 ms frames; aggregate over voiced segments, 1–5 s or turn; normalize to speaker baseline, often in semitones.	Moderate-high	Sex/anatomy, age, language/ lexical tone, codec, pitch-tracking errors.	openSMILE/ eGeMAPS, Praat. 37
F0 range / variability / slope	Dispersion and contour dynamics of pitch.	Frame extraction; turn-level percentiles/range and local slopes.	Moderate	Same as F0 plus phrase type/ question intonation.	openSMILE, Praat.

Dimension / feature	Definition and extraction	Typical realtime window / aggregation	Reliability	Main confounds	Example tools
Intensity / RMS energy	Signal amplitude or estimated sound-pressure-related energy; use relative change more readily than absolute loudness unless input is calibrated.	20–50 ms frames; 1–5 s/turn statistics.	High for digital energy; moderate psychologically	Microphone gain, AGC, distance, codec, device volume.	openSMILE, Praat, librosa RMS. ³⁸
Speaking rate	Words, syllables or phones per unit time including pauses.	Prefer turn or ≥ 5 s window; rolling 10–30 s baseline.	High–moderate	Language, ASR errors, individual baseline, task complexity.	ASR timestamps; syllable detector; openSMILE.
Articulation rate	Speech units per unit of active speech, excluding pauses.	Turn or ≥ 5 s voiced/speech segment.	Moderate–high	VAD threshold and language-specific syllabification.	VAD + ASR / forced alignment.
Pause rate / duration	Count and distribution of within-turn silent intervals.	Detect at roughly 100–500 ms thresholds depending use; aggregate per turn/rolling minute.	High	Packet loss, VAD, deliberate hold, overlap, breath sounds.	Silero/Riva VAD, ASR timestamps. Silero documents lightweight CPU inference; Riva supports neural VAD/endpointing. ³⁹
Spectral/voice-quality features	HNR, spectral slope/tilt, alpha ratio, formant/spectral characteristics and related phonation descriptors.	Voiced frames; aggregate over several seconds/turn.	Moderate	Codec, microphone, room, phonetic content, speaker physiology.	eGeMAPS/openSMILE; Praat; librosa for spectral descriptors. ⁴⁰

Dimension / feature	Definition and extraction	Typical realtime window / aggregation	Reliability	Main confounds	Example tools
Jitter / shimmer	Cycle-to-cycle F0-period and amplitude perturbation.	Clean voiced segments, preferably sustained/stable phonation; offline or selectively.	Low-moderate in natural telephone conversation	Pitch tracking, consonants, noise, codec, algorithm implementation, voice pathology.	Praat/ Parselmouth. Praat notes that implementations can differ because of voicing, period and peak-selection methods. ⁴¹
General SSL embedding	Dense representation from a pretrained speech model capturing phonetic, speaker, channel and paralinguistic structure.	Usually ~1–5 s or per turn; pool frames; store only if needed.	Moderate	Unknown representation factors, channel/domain shift, potential speaker identity leakage.	wav2vec-family/ WavLM-style encoders; SpeechBrain ecosystem.
Emotion-specific embedding	Learned representation optimized or pretrained around emotional/ paralinguistic information.	~2–5 s and/or turn pooling; longitudinal deltas.	Moderate after validation	Emotion-corpus mismatch, acted speech, language and culture.	emotion2vec. ⁶
SER probabilities / valence-arousal	Model posterior or regression output for affect categories/ dimensions.	Prefer 3–10 s/turn; smooth rather than reacting frame-by-frame; preserve probabilities and abstention.	Low-moderate to moderate	Subjective labels, OOD audio, channel, language, speaker demographics, poor calibration.	Fine-tuned emotion2vec/ wav2vec2, SpeechBrain baselines, custom multimodal classifier. ⁴²

Dimension / feature	Definition and extraction	Typical realtime window / aggregation	Reliability	Main confounds	Example tools
Turn duration / count / talk ratio	Time each participant controls the floor and frequency of turns.	Native turn events; aggregate 30–120 s and call.	High	Incorrect speaker/channel assignment, overlap.	VAD + channel identity/diarization; Amazon and other contact-center systems expose related metrics. ⁴³
Turn-switch gap	End of one speaker's speech to beginning of the other's. Negative values indicate overlap.	Every transition; report median/p90/p95 and distribution by direction.	High	Endpoint/VAD errors.	Timestamped VAD/audio. Turn timing is a central spoken-dialogue quality variable. ⁸
Overlap / interruption	Simultaneous speech; interruption is the pragmatically meaningful subset where one party takes or attempts to take the floor.	Event level; rolling rates per minute/call.	High for overlap; moderate for “interruption” semantics	Backchannels, accidental noise, duplex/audio echo, diarization.	pyannote, NeMo/Riva diarization, channel audio; Google CX analytics explicitly measures interruption. ⁴⁴
Backchannel	Short acknowledgment such as “uh-huh,” “right,” “okay” that does not necessarily claim the floor.	Event classification; rate and appropriate-timing metrics.	Moderate	Language/culture; ambiguous short turns; ASR deletion.	Audio+text classifier/LLM. Turn-taking research treats backchannels as a distinct modeling problem. ²⁰
Trajectory / slope	Change of a metric relative to earlier conversation or personal baseline.	EWMA or robust slope over ~30–120 s; start→middle→end.	Depends on source feature; often more useful than absolute value	Topic changes, speaker/channel changes.	Stream processor/time-series engine.

Dimension / feature	Definition and extraction	Typical realtime window / aggregation	Reliability	Main confounds	Example tools
Volatility / change points	Variance, sudden transitions and persistence of sentiment/prosody/interaction state.	30-120 s plus turn-indexed change-point detection.	Moderate	Sparse turns and segmentation errors.	Statistical streaming layer.
Endpointing latency	Time from actual end of user speech to system's end-of-turn decision.	Every user turn; p50/p90/p95/p99.	High	Ground-truth EOT ambiguity.	Media/VAD/ASR timestamps. NVIDIA describes endpointing as critical to timely finalization without premature cutoff. ⁴⁵
Agent response / first-audio latency	User speech end to first audible synthesized-agent sample.	Every system turn; percentile distributions, not just mean.	High	EOT ambiguity, buffering, client/network delay.	Instrument media gateway, model and TTS events. Spoken-dialogue literature identifies runtime latency as a major fluidity constraint. ¹⁵
TTS control state	Voice ID/model, requested speaking rate, pitch/style/expressiveness, volume, language and synthesis configuration.	Log per response and on changes.	High	Delivered audio may differ from requested setting due model behavior.	TTS/provider API + application telemetry. Azure, for example, documents neural/HD voice behavior and expressive adaptation. ⁴⁶

Dimension / feature	Definition and extraction	Typical realtime window / aggregation	Reliability	Main confounds	Example tools
Barge-in stop latency	Customer speech onset to cessation of agent TTS.	Every barge-in; p50/p95/p99.	High	False VAD triggers and echo.	Media gateway + playback controls. Native voice APIs explicitly support interruption-oriented interaction. ⁴⁷
Network QoS	RTT, jitter, packet loss, reconnects and buffer underruns.	1–5 s telemetry + call percentiles.	High	Measurement point and transport topology.	WebRTC/ WebSocket/ media metrics. OpenAI's realtime engineering guidance emphasizes RTT, jitter and packet loss as important to stable voice interaction. ⁴⁸
Tool/model latency and errors	LLM time-to-decision, tool-call duration, retries, failed actions, TTS generation latency.	Every operation; span-level tracing and call-level percentiles.	High	Async operations and missing trace propagation.	OpenTelemetry/ application traces; provider request telemetry.

A crucial technical distinction is **absolute versus within-speaker features**. An F0 of 180 Hz should not be interpreted as intrinsically “more aroused” than 120 Hz across arbitrary speakers. Track robust within-speaker or within-session baselines, such as median F0 over initial neutral-enough speech, and express subsequent deviations in semitones/z-scores where appropriate. Demographic variation, linguistic tone systems and context all alter the acoustic realization of social and affective information. ⁴⁹

Channel normalization is equally important. In one controlled codec study, all thirteen tested prosodic parameters showed significant codec-specific distortion relative to WAV, with mean measurement deviations reaching approximately 20% in some conditions. Consequently, store codec, sample rate, device/channel type, packet-loss statistics and any gain-control information alongside acoustic observations. ¹⁶

Mapping signals to psychological and behavioral indicators

The evidence levels below mean: **stronger** = systematic/repeated evidence and/or naturalistic customer-service evidence; **moderate** = meaningful evidence but context dependence; **weak** = indirect or insufficiently validated for production inference.

Target indicator	Useful evidence combination	Evidence strength	Safe interpretation
Arousal / activation	Within-speaker F0 level/ range, intensity, speech rate, temporal dynamics; optionally acoustic embeddings.	Moderate-strong for relative activation; weak for cause. Reviews find recurring prosodic relationships, but the same changes occur in multiple states. ¹⁸	"Vocal activation increased relative to this speaker's baseline," not "the customer is angry."
Frustration / negative affect	Negative/complaint language + repetition/ repair + elevated activation + interruption/ shorter gaps + unsuccessful system actions + negative trajectory.	Moderate-strong multimodally. Real customer-service work supports linguistic + acoustic turn-level detection. ⁵⁰	Best production emotion target because it can be operationally anchored to complaint behavior and failures.
Anger	Negative semantics plus elevated F0/intensity/rate and sustained trajectory.	Moderate. These acoustic relationships recur in the 2025 review, but are not individually diagnostic. ¹⁸	Emit probability/ evidence score and confidence, never a categorical "customer is angry" fact.
Stress	F0/intensity changes, duration/rate patterns, lexical context and task difficulty.	Moderate at population level; lower individually. ¹⁸	Appropriate for research/aggregate adaptation only after domain validation; not clinical assessment.
Fear / anxiety	Multimodal semantic/ contextual evidence, possibly prosody.	Weak acoustically. The recent systematic review found no consistent acoustic pattern for fear or anxiety. ¹⁸	Do not deploy an acoustic-only anxiety detector.

Target indicator	Useful evidence combination	Evidence strength	Safe interpretation
Engagement	Timely reciprocal turns, relevant follow-up, backchannels, low unexplained silence, semantic continuation, voluntary elaboration and possibly acoustic entrainment.	Moderate/indirect. Speech-rate/prosodic entrainment occurs in conversation and has been associated with social/communicative variables, but it is not a unique engagement biomarker. ⁵¹	Better framed as “interactional engagement signals” than a latent mental-state score.
Trust	Direct customer rating plus behavioral evidence: following instructions, successful authentication/action, low repair, consistency, resolution; voice features only as supplementary predictors.	Moderate for perceived vocal trustworthiness, weak for individual actual trust. Isolated acoustic cues are inconclusive and context/demographics matter. ¹⁹	Measure actual trust using survey/behavioral criterion; don't infer it from pitch.
Confidence / uncertainty	Hedges, pauses, corrections, duration and pitch/intensity patterns.	Moderate but language/context specific. Experimental evidence demonstrates prosodic and segmental encoding of intended confidence, with lexical-tone interactions. ³⁶	Useful for “possible uncertainty/need for clarification,” not truthfulness.
Customer satisfaction	Direct CSAT or other outcome + first-contact resolution + repeat-contact behavior + effort/repair + sentiment trajectory + interaction quality.	Moderate-strong when directly supervised on satisfaction; inadequate if sentiment is used as a surrogate. Speech/customer-service studies support multimodal forecasting. ⁵²	Train directly against the business construct you care about.
Empathy / rapport	Acknowledgment/validation language, appropriate timing, low harmful interruption, responsive reformulation and possibly entrainment.	Moderate for communication behavior; weaker for internal “empathy.” Genesys itself distinguishes transcript-based empathy analytics; entrainment literature supports social associations. ⁵³	Label observable empathic communication behaviors rather than an agent/customer personality trait.

Target indicator	Useful evidence combination	Evidence strength	Safe interpretation
Cognitive load	Repairs, hesitation, longer pauses, semantic difficulty, F0/intensity and failure/retry context.	Moderate but nonspecific. Systematic evidence finds some F0/intensity association but also ambiguity. ¹⁸	Useful as a “conversation difficulty” signal when combined with observable task complexity.
Compliance likelihood	Historical behavioral outcomes, task context, explicit objections/acceptance, eligibility and agent action—not acoustic cues alone.	Weak as a psychological voice inference. I found insufficient customer-service evidence to regard acoustic-only compliance inference as validated; adjacent trust/confidence literature is highly context sensitive. ⁵⁴	Reframe as a task-specific, ethically reviewed behavioral propensity model when genuinely needed; do not optimize covert persuasion from presumed emotional vulnerability.
Deception / personality / mental-health diagnosis	—	Not recommended for this use case. Speech models exhibit demographic variation and state/trait ambiguity; acoustic correlates are nonspecific. ⁵⁵	Exclude from ordinary customer-service production analytics.

For an adaptive agent, an especially useful state representation is therefore not:

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emotion = "anger"
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but something closer to:

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negative_semantic_valence = 0.81
vocal_activation_delta = +1.4 SD
repair_rate_60s = 3
interruption_rate_60s = 2
unresolved_tool_failures = 1
frustration_probability = 0.72
frustration_calibration_bin_accuracy = 0.67
confidence = "moderate"
```

This makes the underlying evidence auditable and permits the policy layer to distinguish “negative wording with calm prosody” from “rapid escalation following three failed authentication attempts.”

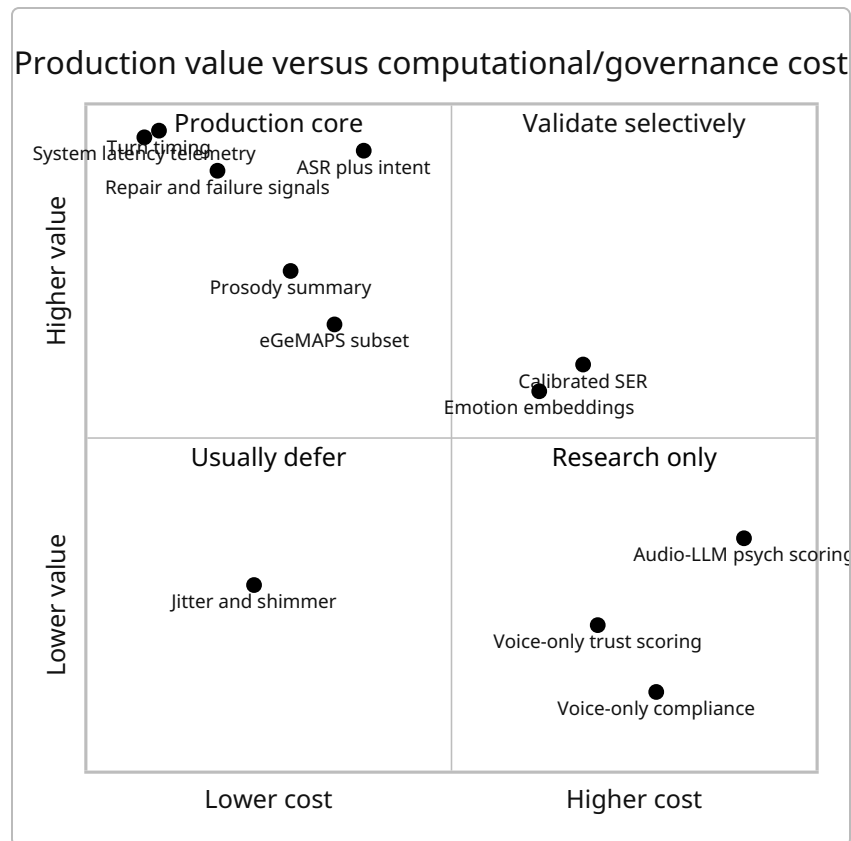
Production priorities, privacy, and compliance

The tiers below reflect **customer-service production value, measurement reliability, latency/cost and governance burden**, not merely predictive feature importance.

Tier	Feature families	Rationale	Relative compute	Privacy / compliance
Essential	Separate customer/agent channels where technically possible; reliable clocks/timestamps; audio quality/codec metadata; VAD; transcript + word/turn timing; speaker role; silence; turn gaps; overlap; barge-in; talk time; intent/topic; repair/repetition; tool outcome/errors; first-audio/endpointing/tool/TTS/network latency; actual resolution/escalation/repeat-contact/CSAT.	These are observable, explainable and directly actionable. Mainstream contact-center analytics already center many of them, and spoken-dialogue research shows timing is integral to interaction quality. ⁵⁶	Low except STT	Transcript/raw audio can contain extensive PII. Redact/minimize, enforce retention/access policies and separate identifiers from analytic stores. ⁵⁷
Recommended	Speaker-normalized F0, range, intensity, speech/articulation rate, pause features; compact eGeMAPS; lexical sentiment; probabilistic valence/arousal/negative affect; interaction trajectories; backchannels; learned embeddings where justified; agent TTS control logs.	Adds useful evidence for frustration and conversation dynamics while retaining some interpretability. GeMAPS and recent systematic evidence support this feature family. ⁵⁸	Low-moderate; embeddings moderate/GPU-friendly	Acoustic features can carry demographic/speaker information. Avoid retaining more dimensions/precision than operationally required and test for subgroup leakage/bias. ¹⁷

Tier	Feature families	Rationale	Relative compute	Privacy / compliance
Exploratory	Jitter/shimmer in connected telephony; high-dimensional embeddings at high frame rates; generic audio-LLM emotion labels; trust/confidence/compliance estimators; interpersonal entrainment as a quality score.	Scientifically interesting but more sensitive to domain/channel/label ambiguity; value should be demonstrated experimentally before operational use. ⁵⁹	Moderate-high	High-dimensional voice representations can increase identifiability and model-governance burden.
Generally avoid	Deception, personality, psychiatric-condition inference, protected-class inference, employee emotional-state ranking, adverse customer decisions based on inferred emotion.	Insufficient specificity and elevated bias/rights risks. ⁵⁵	Any	Potentially severe legal, employment, discrimination and biometric/privacy exposure; European workplace emotion inference receives especially restrictive treatment. ⁶⁰

The chart below is a qualitative engineering prioritization, not a published feature-importance meta-analysis.



Privacy and legal design constraints

Recordings and transcripts should be considered highly sensitive application data even before biometrics enter the picture. They may contain account information, payment data, health information, names, addresses, authentication material and free-form disclosures. Major contact-center products therefore incorporate PII/sensitive-data redaction as a first-class capability, and governance frameworks such as NIST AI RMF and ISO/IEC 42001 emphasize lifecycle risk management rather than merely model accuracy. ⁶¹

Voice analytics can cross into biometric regulation. Illinois BIPA expressly includes a “voiceprint” in its definition of biometric identifier. This does not mean every RMS-energy or pitch measurement is automatically a regulated voiceprint, but it is a strong reason not to retain speaker-recognition embeddings or build identity-capable voice representations unless the business need and applicable legal basis have been explicitly reviewed. ⁶²

Recording and monitoring consent should be resolved jurisdiction by jurisdiction before capture/analysis begins. U.S. federal wiretap law prohibits unauthorized interception subject to statutory exceptions, including consent-related exceptions; production policy should not assume that satisfying a federal baseline resolves every other applicable jurisdictional requirement. ⁶³ Maintain a machine-readable consent/legal-basis record attached to the session, including what was disclosed, when, jurisdiction/policy version, allowed analytics and retention class.

The EU AI Act creates an important distinction between simple acoustic observation and emotion inference. Its definition concerns systems intended to identify/infer emotions or intentions from biometric data; its recitals explicitly distinguish mere observation of readily apparent behavior such as a raised voice or whisper unless those observations are used to infer emotion. The Act prohibits emotion-inference systems in workplace and educational settings subject to specified medical/safety exceptions and requires notification of persons exposed to permitted emotion-recognition systems. ⁶⁴ For a contact center, this makes **“customer speaking-rate increased by 25%”** legally and scientifically a materially different design from **“employee is angry and receives a poor performance score.”**

For U.S. **outbound** automated voice agents, the FCC has confirmed that AI-generated humanlike voices fall within the TCPA’s “artificial or prerecorded voice” restrictions; the FCC’s ruling states that such calls require prior express consent absent an applicable emergency purpose or exemption. ⁶⁵ This is separate from recording/analytics consent and should be treated as a distinct compliance control.

A sensible retention hierarchy is therefore:

lowest retention sensitivity: aggregate operational metrics → low-dimensional non-identifying acoustic statistics → transcript after PII redaction → high-dimensional embeddings → raw audio / speaker-recognition material.

The exact classification depends on purpose, jurisdiction and implementation, but the architectural principle of **purpose limitation and data minimization** is consistent with modern AI-risk-management practice. ⁶⁶

Evaluation and operational monitoring

A model that achieves a good emotion F1 score on an acted corpus is not sufficiently validated for customer service. Naturalistic customer-care work explicitly identifies realistic-data scarcity and acted/naturalistic performance gaps, while research on speech-based ML documents differential feature distributions and performance across demographic groups. ⁶⁷

The evaluation program should therefore operate at four different levels.

Ground truth and labeling

Use different truth sources for different constructs rather than forcing them into one generic “sentiment” label.

Construct	Preferred ground truth
Speech/timing mechanics	Human-corrected words, boundaries, speaker/channel identity, overlap events and timestamps on a stratified audit set.
Observable dialog behavior	Trained annotators marking interruption, acknowledgment, repair, refusal, escalation, explicit complaint, resolution, etc.

Construct	Preferred ground truth
Emotion / frustration	Multiple independent raters plus, where feasible, customer self-report and context. Preserve label distributions/disagreement rather than pretending every ambiguous turn has one objectively true emotion. Speech-emotion uncertainty is a documented deployment problem. ⁶⁸
Trust	Direct customer survey and/or prespecified behavioral criterion, not an acoustic proxy. Trustworthiness literature specifically warns against isolated-feature interpretations. ¹⁹
Satisfaction	CSAT/NPS-like survey where available, first-contact resolution, repeat contact, complaint/escalation, task success and related explicit outcomes.
Agent quality	A rubric of observable behaviors such as correct disclosure, acknowledgment, verification, resolution procedure and prohibited behavior—not inferred emotional disposition.
System quality	Latency traces, errors, retries, WER audit, endpointing error, false interruption, tool outcome and audio/network telemetry.

For subjective annotations, start with a precise labeling manual, train annotators on edge cases and measure **Krippendorff's α or weighted κ for categorical/ordinal judgments and ICC for continuous ratings**, with bootstrap confidence intervals where practical. Rather than hiding disagreement, retain rater distributions for ambiguous emotion labels; contemporary work on speech uncertainty highlights that emotion labels themselves are an important uncertainty source. ⁶⁹

At least one holdout should be **time-separated**, and splits should prevent the same speakers/customers from leaking across training and test sets when speaker-specific acoustic characteristics could otherwise be memorized. Also stratify evaluation by channel/codec, SNR, locale/language/accent, call type, duration and legally appropriate demographic groups. Cross-domain naturalism and demographic bias research make this substantially more important for speech than random utterance-level splits suggest. ⁷⁰

Model metrics

Do not optimize a single headline accuracy.

Component	Core offline metrics	Operational additions
ASR	WER/CER; entity/name error; deletion rate; word-timestamp error	Partial/final latency, endpoint delay, correction frequency; WER by locale/channel. Spoken-interface research recommends evaluating timing, overlap, diarization and coverage along with WER. ¹⁵
VAD / endpointing	Speech precision/recall/F1; boundary error	Premature endpoint rate, excessive-wait rate, false barge-in events/hour.

Component	Core offline metrics	Operational additions
Diarization	DER/JER; speaker confusion	Role-assignment failure, overlap-specific DER, downstream analytics error attributable to diarization.
Categorical emotion/frustration	Macro-F1, unweighted average recall, per-class precision/recall; AUROC/AUPRC for one-vs-rest where useful	False escalations/hour, abstention/coverage, calibration by subgroup/channel.
Continuous valence/arousal	MAE/RMSE; Pearson/Spearman; concordance correlation coefficient	Error by intensity, channel and interaction phase.
Probability quality	Brier score, log loss/NLL, reliability diagrams, ECE alongside more detailed bin diagnostics	Calibration drift, risk-versus-coverage under abstention. Speech-classification research specifically reports overconfidence/calibration problems. ⁷¹
Turn taking	EOT accuracy, gap error, overlap detection	User→agent gap distribution; premature interruption rate; barge-in handling; abandoned user turns.
End-to-end agent	Task success, grounded correctness, safety/QA rubric	Resolution, escalation, repeat contact, containment, customer effort/satisfaction, latency and complaint guardrails.

For rare but expensive events—for example, truly severe frustration leading to escalation—**precision-recall curves and expected business cost** are more useful than raw accuracy. A model that calls everyone “not severely frustrated” can have excellent accuracy on an imbalanced dataset while being useless.

Calibration deserves independent treatment. A nominal 0.8 frustration probability should correspond to something operationally meaningful in the validation population; current speech-classification research warns that neural classifiers can be overconfident, and real-world SER contains uncertainty from ambiguous labels, OOD input and corrupted signals. ²⁴ Use temperature/vector calibration or another held-out calibration method as appropriate, and implement an **abstain/unknown** state rather than forcing a prediction on silence, noise or clear domain shift.

A/B and causal evaluation

When an inferred state changes the agent's behavior, evaluate the **adaptation policy**, not just the detector.

For example:

Hypothesis:
Customers exhibiting convergent frustration evidence benefit from

shorter, acknowledgement-first responses and earlier human escalation.

Control:

Normal agent policy.

Treatment:

Same base model, but policy receives calibrated frustration evidence.

Primary endpoint:

Task-resolution or repeat-contact outcome.

Secondary:

CSAT / escalation quality / abandonment.

Guardrails:

Latency, inappropriate escalation, false apology, interruptions, safety incidents, complaint rate.

Analysis:

Intention-to-treat at customer/session level;
prespecified subgroup and channel checks.

Session- or customer-level randomization is generally preferable to turn-level randomization because a treatment response changes subsequent conversation state, creating interference between turns. Run separate experiments for TTS pacing, interruption policy, acknowledgment strategy and escalation thresholds so that improvements can be attributed to specific mechanisms.

Most importantly, compare adaptive policy against **observable-event baselines**. For example, test whether a sophisticated frustration model adds incremental value over simple variables such as repeated failure + explicit complaint + interruption + three repairs. If it does not, the simpler model will usually be easier to defend and maintain.

Drift monitoring

Monitor at least four drift layers:

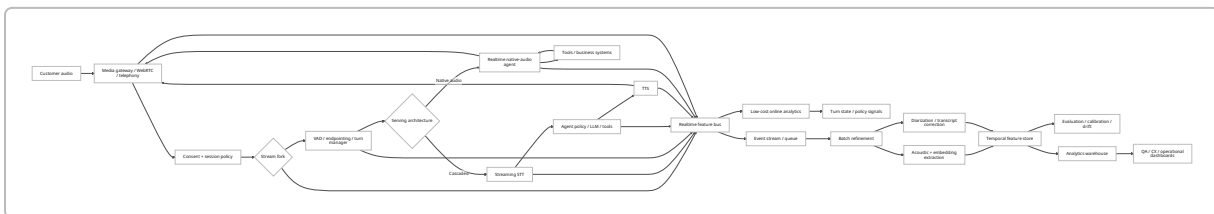
Layer	Signals
Input/channel drift	Sample rate, codec, SNR proxies, packet loss, duration, VAD ratio, acoustic feature distributions. Codec studies demonstrate that channel changes can alter prosodic measurements themselves. ¹⁶
Population/content drift	Language/locale, topic, intent, vocabulary, embedding distributions, new product names/issues.
Model drift	WER audits, confusion matrices, Brier/NLL/ECE, abstention rate, feature-attribution changes.

Layer	Signals
Outcome drift	Resolution, repeat contact, escalation, abandonment, CSAT and false-policy-trigger rates.

Use rolling baselines, distributional tests or divergence measures for alerting, but trigger **human review and labeled re-evaluation** rather than automatic retraining solely because a statistical threshold moved. NIST's AI RMF framework explicitly organizes AI risk work around ongoing Govern, Map, Measure and Manage activities. ⁷²

Implementation architecture, latency, and data model

The architecture should keep **conversation serving** and **measurement/evaluation** logically separable. This remains true when using a native-audio model.



The value of the feature bus is that **the serving model is no longer your only observer**. Even if a native-audio model internally hears pace and intonation, your system independently knows: packet loss was 3.7%; customer speech began at 12.422 s; TTS actually stopped at 12.591 s; the last three turns contained two clarification requests; F0 rose relative to baseline; and the authentication tool failed twice. This produces a reproducible record of why the policy adapted.

Native-audio APIs are increasingly capable of collapsing the traditional pipeline. Google describes Gemini Live as processing continuous audio/video/text and supporting interruption; Microsoft Voice Live provides a combined recognition-generative-AI-TTS interface; OpenAI's realtime architecture similarly emphasizes direct audio interaction. ⁷³ The tradeoff is that explicit intermediate representations may be less available, making **external observability more important, not less**.

Realtime versus batch placement

Capability	Realtime path	Offline/refinement path
VAD / speech onset	Yes	Recompute for audit if necessary
End-of-turn decision	Yes, critical	Evaluate boundary errors
ASR partials	Yes	—
Final/corrected transcript	Turn-final	Yes , optionally higher-accuracy model
Turn timing / overlap	Yes	Refine with diarization

Capability	Realtime path	Offline/refinement path
Intent / explicit complaint	Yes	Whole-call reclassification
Lightweight sentiment	Yes	Contextual whole-call scoring
F0 / intensity	Yes	Re-extract with standardized pipeline
Rate / pause metrics	Incremental	Yes , more reliable on complete turns
eGeMAPS	Compact subset if needed	Recommended full feature vector
SSL/emotion embeddings	Only if low-latency GPU capacity justified	Recommended location
SER	Turn-level if policy demonstrably benefits	Recalibration and full trajectory
Jitter / shimmer	Usually no	Selective research/QA
Audio-capable LLM retrospective analysis	Only when part of serving model	Useful for QA/research, not sole measurement
Conversation trajectories	Rolling simplified form	Yes
QA / outcome modeling	Minimal	Yes

Latency targets

Human-human turn transitions are remarkably fast; a 2025 turn-taking survey cites an average roughly 200 ms silence between human utterances and notes substantially larger historical gaps in spoken agents, while the timing-bottleneck literature identifies ASR runtime latency as a barrier to fluid conversation. ⁸ These numbers should not be turned into a universal 200 ms SLA: some customer-service interactions benefit from deliberate pauses, and end-of-turn ambiguity creates a speed-versus-interruption tradeoff.

For an unspecified-scale system, I would begin with these **engineering objectives**, then tune from user experiments rather than treating them as standards:

Measure	Initial target
Audio processing frame	20–40 ms
VAD compute per incoming chunk	comfortably below realtime; ideally single-digit ms on target hardware
Speech-onset → barge-in stop	p95 ≤ 150 ms
Incremental transcript update after received speech	p95 ≤ 200–300 ms , provider/network dependent

Measure	Initial target
Definite user EOT → lightweight analytics update	≤250 ms
User EOT → first audible agent response	p50 ≤500 ms; p95 ≤800 ms for ordinary non-tool turns
Long tool-call response	Stream acknowledgment/backchannel rather than allowing silent multi-second waits
Monitoring	p50/p90/p95/p99 by geography, client type, call provider and model version

These are design targets rather than claims about any vendor SLA. The goal is to make the **tail distribution** visible. A 400 ms mean hides a system where 10% of turns take two seconds.

Streaming VAD can be extremely inexpensive: Silero's published implementation describes sub-millisecond processing of a roughly 30 ms audio chunk on a single CPU thread under its benchmark conditions. ⁷⁴ The expensive components are much more likely to be large STT/audio encoders, multimodal foundation models and TTS rather than deterministic timing/prosody calculations.

Storage and compute estimates

The estimates below are deliberately order-of-magnitude. They assume numeric features stored compactly in a columnar/binary representation; JSON can multiply storage due to field names/timestamps. Audio baseline is **16 kHz, 16-bit PCM mono**, for which one hour is approximately **115.2 MB per channel**. A dual-channel uncompressed recording is therefore approximately 230.4 MB/h before container overhead.

Feature	Type	Example extractor	Window / hop	Approx. stored data per hour	Compute
Raw PCM audio	int16	Media layer	Continuous	115 MB/channel	Negligible
VAD regions	event intervals	Silero / Riva	20–50 ms detection; store events	~0.01–0.05 MB	Very low
Word transcript + timestamps	text/events	Whisper / Azure / Riva	Word/segment	~0.3–1 MB	High relative to DSP
Turn metadata	events	VAD + speaker role	Turn	~0.02–0.1 MB	Very low

Feature	Type	Example extractor	Window / hop	Approx. stored data per hour	Compute
Sentiment/ intent/ dialogue labels	probabilities/ events	Small classifier / LLM	Turn	~0.05–0.3 MB	Low–high depending model
F0/intensity/ rate/pause summary	~8 float32	openSMILE / Praat + timestamps	1 s summaries	~0.12 MB numeric; ~0.2–0.5 MB with metadata	Low
Higher-rate prosody stream	~25 float32	openSMILE	100 ms hop	~3.6 MB numeric	Low
eGeMAPS	88 float32	openSMILE	5 s hop	~0.25 MB numeric; ~0.4–0.8 MB practical	Low
Voice-quality summary	float vector	openSMILE/Praat	5 s / turn	~0.1–0.3 MB	Low
Emotion probability vector	e.g. 8 float32	SER model	5 s hop	~0.023 MB numeric; ~0.05–0.2 MB practical	Moderate
768-D acoustic embedding	float16[768]	emotion2vec/SSL encoder	5 s hop	~1.1 MB	Moderate–high
Same 768-D embedding	float16[768]	SSL encoder	1 s hop	~5.5 MB	Moderate–high
Network/QoS telemetry	numeric events	WebRTC/media stack	1–5 s	~0.05–0.5 MB	Negligible
Model/tool trace events	spans/events	OpenTelemetry/ app logs	Request/ event	~0.1–2+ MB	Negligible extraction; storage depends on verbosity
Native-audio LLM retrospective labels	structured text	Audio-capable LLM	Turn/call	Small result storage	High inference cost

The storage arithmetic makes an important point: **interpretable analytics are cheap to store**. A rich turn/prosody/eGeMAPS stream can consume far less than one percent of raw PCM storage. High-rate embeddings grow faster and are much more difficult to interpret or govern, so “we can store them” is not a sufficient reason to do so.

Example event schema

A production schema should keep observations, inferences and outcomes explicitly distinct:

```
{
  "schema_version": "1.0",
  "session": {
    "session_id": "01J...",
    "started_at": "2026-08-08T14:03:11.423Z",
    "channel": "webrtc",
    "locale_hint": "en-US",
    "consent": {
      "recording_allowed": true,
      "analytics_allowed": true,
      "emotion_inference_allowed": false,
      "policy_version": "consent-us-2026-07"
    }
  },
  "audio_context": {
    "sample_rate_hz": 16000,
    "codec": "opus",
    "channel_role": "customer",
    "packet_loss_pct": 0.3,
    "rtt_ms": 46,
    "snr_proxy_db": 21.8
  },
  "turn": {
    "turn_id": "t-0042",
    "speaker_role": "customer",
    "start_ms": 82411,
    "end_ms": 87192,
    "asr": {
      "text": "I've already tried that twice.",
      "mean_confidence": 0.93,
      "word_count": 6,
      "model": "stt-model-version"
    }
  },
  "lexical": {
```

```

    "sentiment": {
      "negative": 0.83,
      "neutral": 0.15,
      "positive": 0.02
    },
    "dialogue_act": "correction",
    "repair": true,
    "explicit_complaint": true,
    "intent": "technical_support"
  },

  "acoustic_observations": {
    "f0_median_hz": 181.4,
    "f0_delta_semitones_from_session_baseline": 2.1,
    "f0_iqr_semitones": 4.2,
    "rms_relative_db": 3.6,
    "speech_rate_words_per_sec": 3.4,
    "pause_fraction": 0.04
  },

  "interaction_observations": {
    "preceding_gap_ms": 138,
    "overlap_ms": 212,
    "customer_interrupted_agent": true,
    "repairs_last_60s": 3,
    "tool_failures_last_60s": 2
  },

  "behavioral_inferences": {
    "negative_affect_probability": 0.78,
    "frustration_probability": 0.84,
    "valence": -0.63,
    "arousal": 0.71,
    "abstained": false,
    "model": "frustration-mm-v7",
    "calibration_version": "cal-2026-07-en-us"
  },

  "system_response": {
    "endpointing_latency_ms": 241,
    "policy_latency_ms": 187,
    "tool_latency_ms": 0,
    "tts_first_audio_latency_ms": 198,
    "user_end_to_agent_audio_ms": 626,
    "barge_in_stop_ms": null,
    "tts_voice": "support-neutral-2",
    "tts_rate_multiplier": 0.96,
    "response_strategy": "acknowledge_and_escalate"
  }

```

```
    }
  },

  "privacy": {
    "raw_audio_retention_class": "short",
    "transcript_pii_redacted": true,
    "speaker_identity_embedding_stored": false
  }
}
```

Notice that `frustration_probability` does **not** overwrite the constituent measurements. That design lets you change the inference model later, recompute it offline and audit whether a policy decision made under `frustration-mm-v7` would still occur under `v8`.

Tool and model landscape

There is no single best stack because a fully managed contact-center system, a self-hosted GPU deployment and a low-volume SaaS agent have radically different constraints. The table therefore separates **measurement infrastructure** from **native conversational models**.

Category	Tool / model	Strengths	Limitations	Recommended use
STT	OpenAI Whisper	Open model; robust multilingual general-purpose transcription; trained on a large diverse web speech corpus; widely supported ecosystem. ⁷⁵	Original architecture is naturally chunk/ batch oriented rather than a complete low-latency conversation stack; streaming often comes from wrappers/ derivatives.	Offline/reference transcription, prototyping, self-hosted pipelines.
Realtime STT / voice	OpenAI realtime speech models, including GPT-Realtime-Whisper / GPT-Realtime-2 family documented in 2026	Direct realtime voice stack and streaming transcription; integrates naturally with voice agents. ⁷⁶	Managed service; internal acoustic representations are not an analytics API; model/version behavior must be monitored.	Low-latency serving plus an independent analytics plane.

Category	Tool / model	Strengths	Limitations	Recommended use
STT	Azure Speech	Real-time, fast and batch transcription; locale coverage and custom-speech capabilities. ⁷⁷	Cloud dependency unless using supported container/ embedded options; performance is locale/domain dependent.	Enterprise STT where Azure integration/ governance is advantageous.
STT + diarization	NVIDIA Riva / Speech NIM	GPU-oriented deployment; streaming ASR; streaming speaker diarization; endpointing; published performance benchmarking. ⁷⁸	GPU infrastructure/ operations; feature maturity and concurrency limits can vary by release. ⁷⁹	High-scale controlled infrastructure/ on-prem/VPC deployments.
Diarization	pyannote.audio	Mature open-source PyTorch diarization ecosystem with pretrained pipelines and customization options. ⁸⁰	Strong offline use but compute-intensive; overlapping speech and domain mismatch remain hard.	Batch/ refinement diarization; evaluation reference pipeline.
Diarization	NeMo / Sortformer	NVIDIA supports offline/streaming-oriented diarization approaches, including Sortformer. ⁸¹	More operational complexity than a managed API.	GPU stacks needing custom diarization and low-level control.
VAD	Silero VAD	Very small model and extremely low reported CPU latency for short chunks. ⁷⁴	Needs threshold tuning for your noise/channel; VAD alone is not semantic EOT prediction.	Default open-source VAD candidate for realtime pipelines.

Category	Tool / model	Strengths	Limitations	Recommended use
VAD / endpointing	Riva	Integrates neural VAD and end-of-utterance handling with the streaming speech stack. ⁸²	Couples feature to NVIDIA deployment stack.	Integrated enterprise GPU voice stack.
Acoustic features	openSMILE / eGeMAPS	Standardized, highly interpretable affective/prosodic feature extraction. Current docs expose eGeMAPS feature sets and the much larger ComParE set. ⁸³	Handcrafted features cannot capture all relevant paralinguistic structure; many are correlated.	Recommended default acoustic extractor for QA and explainability.
Acoustic/voice quality	Praat / Parselmouth	Detailed F0, intensity, formant, jitter, shimmer, HNR and phonetic analysis. ⁸⁴	Some measurements are fragile in connected/noisy/codec speech; implementation settings matter. ⁴¹	Offline research, validation and specialized voice-quality analysis.
General DSP	librosa	Convenient MFCC, RMS, spectral centroid/ bandwidth and other DSP primitives. ⁸⁵	Does not by itself provide an affectively validated feature ontology.	Custom feature engineering and signal-quality analytics.
Emotion representation	emotion2vec	Purpose-built self-supervised emotion representation; reported cross-language gains across ten languages and multiple affect-related tasks. ⁶	Research model; production accuracy must be established on naturalistic contact-center data.	Strong candidate for experimental embedding layer/fine-tuned SER.

Category	Tool / model	Strengths	Limitations	Recommended use
Emotion baseline	SpeechBrain wav2vec2/ IEMOCAP models	Easy-to-use pretrained baseline and reproducible open tooling. ⁸⁶	Trained/evaluated on IEMOCAP rather than your customer-service distribution; acted/ corpus transfer is a known concern. ²¹	Benchmark only; do not deploy unchanged as ground truth.
Native audio agent	Google Gemini Live / 2.5 Flash Native Audio	GA native-audio architecture documented for realtime voice/ video, interruptions and low-latency interaction; configurable affective dialogue is documented. ⁸⁷	Provider-controlled internal representation; vendor capability to perceive acoustic cues does not mean its implicit emotion judgments are calibrated for your business.	Native conversational serving; supplement with independent measurement.
Native/ integrated voice agent	Azure Voice Live	Single interface joining speech recognition, generative AI and TTS; designed for low-latency natural voice use cases including contact centers. ⁸⁸	Managed architecture and model-specific behavior; less low-level observability than fully DIY pipeline unless instrumented externally.	Azure-centric production voice agents.
Managed contact analytics	Amazon Connect Contact Lens	Turn sentiment, talk/non-talk, loudness, issues and PII workflows integrated with contact-center infrastructure. ⁸⁹	Vendor-defined scoring; less flexible than custom research pipeline.	Operational analytics and QA when already using Connect.

Category	Tool / model	Strengths	Limitations	Recommended use
Managed CX analytics	Google CX Insights / Quality AI	Sentiment, silence, interruption, entities, intent and QA workflow. ⁹⁰	Managed ontology/model behavior.	Large-scale QA/ analytics in Google ecosystem.
Managed CCaaS analytics	Genesys Speech/ Text Analytics	Broad transcription, sentiment, topic/ category and empathy-related workflows. ²⁷	“Empathy” and similar labels need scrutiny to understand whether they measure wording, audio or inferred state.	Operational QA for Genesys deployments.

The most defensible stack for a new system is therefore a **hybrid**:

```

Serving layer
  Native-audio agent
    OR
  streaming STT → policy/LLM/tools → TTS

Independent measurement layer
  VAD + precise timestamps
  + ASR transcript/word timing
  + low-dimensional prosody
  + interaction events
  + system/network/tool telemetry

Asynchronous intelligence layer
  transcript refinement
  + diarization refinement
  + eGeMAPS
  + acoustic embeddings
  + calibrated multimodal affect model
  + conversation trajectory
  + QA/outcome model

Evaluation layer
  human labels
  + customer self-report
  + business outcomes

```

- + calibration/fairness/drift
- + causal A/B experiments

That division aligns well with both the scientific evidence and the evolution of commercial voice platforms. The realtime agent gets the benefits of native audio or fast streaming inference, while the organization retains a stable, inspectable measurement substrate that remains comparable when the conversational model is upgraded. Recent spoken-dialogue research strongly supports treating timing and overlap as first-class phenomena, recent customer-service emotion work supports acoustic-plus-linguistic fusion, and contemporary governance frameworks require ongoing measurement rather than one-time benchmark certification. ⁹¹

The resulting conceptual hierarchy is:

Level A — facts: who spoke, what words were recognized, when speech occurred, whether speakers overlapped, how long the system waited, whether a tool failed.

Level B — derived behavior: repeated clarification, negative lexical trajectory, faster speech than the customer's baseline, increased vocal activation, repeated agent interruption.

Level C — probabilistic interpretation: likely frustration, possible uncertainty, declining engagement.

Level D — psychological/business construct: trust, satisfaction, willingness to comply.

Production decisions should become progressively more conservative from Level A to Level D. The scientific literature is strongest when measuring **speech and interaction**, weaker when inferring **affective state**, and substantially more context-dependent when inferring broad **psychological traits or intentions**. ⁹²

For a customer-service agent specifically, the highest-return optimization loop is likely to be **observable failure → conversation-dynamics change → calibrated frustration evidence → adaptive response → actual outcome**, rather than an unconstrained general-purpose “psychological profiler.” This design is more measurable, more scientifically defensible, easier to evaluate causally, and more compatible with the increasingly restrictive governance environment for emotion and biometric inference. ⁹³

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